



**HINDUSTAN PETROLEUM CORPORATION LIMITED**  
**(A Government of India Enterprise)**  
**QUALITY CONTROL LABORATORY**  
MAZGAON LUBE PLANT  
HAY BUNDER ROAD, MAZGAON  
MUMBAI - 400033, MAHARASHTRA  
**LABORATORY TEST REPORT**



|                              |                                               |                             |                          |
|------------------------------|-----------------------------------------------|-----------------------------|--------------------------|
| Product                      | 2376211 ( HP TRANSFORMER OIL - DRUM 210L EC ) | Test Report Number          | 4546232401205            |
| Loc/ Customer/ Supplier      | 4546 ( MAZGAON TML LUBES )                    | Sample Number               | 40000002393              |
| Process/Purchase/Sales Order | 10004333 4901900419 0001                      | Lab Sr. No                  | FB 522                   |
| Source of the Sample         | DRUM                                          | ULR Number                  | TC-5177230000001205F     |
| Tank No.                     | WHS1 (Lube Warehouse 1)                       | Date of Sampling            | 25.08.2023 15:22:46      |
| Type of the Sample           | COMPOSITE                                     | Date of Receipt of Sample   | 16.08.2023               |
| Number/Quantity of Sample    | 1*1 L                                         | Date(s) of Sample Testing   | 16.08.2023 To 23.08.2023 |
| Batch No.                    | M1-23G0078                                    | Date of Issue of the Report | 25.08.2023 17:08:12      |
| Stock Represented            | 1,167.000 EA                                  | Sample Drawn By             | KUMAR AJAY               |
| Test Required                | FB                                            |                             |                          |
| Sampling Plan/Method         | IS 6855                                       | Seal No.                    | Wooden Box Seal          |
| Reason for Testing           | PRODUCTION - DRUM                             |                             | Aluminium Box Seal       |

| S.No | Characteristics                         | UOM                | Test Method        | Specifications   | Result           |
|------|-----------------------------------------|--------------------|--------------------|------------------|------------------|
| 1    | FLASH POINT, COC                        | °C                 | ASTM D92           | 135.0 - 300.0    | 166.0            |
| 2    | 2-FURURAL AND RELATED COMPOUNDS CONTENT | mg/kg              | IS 15668           | MAX 0.05         | 0.00             |
| 3    | TENDENCY TO ABSORB GAS                  | -                  | IEC 60628 METHOD A | MAX 999.90       | 9.50             |
| 4    | OXIDATION STABILITY TEST                | mgKOH/g            | JIS K 2514         | MAX 1.20         | 0.00             |
| 5    | DIELECTRIC DISSIPATION FACTOR OXIDISED  | -                  | IS 12422           | MAX 0.500        | 0.002            |
| 6    | SLUDGE OXIDISED                         | %Wt                | IS 12422           | MAX 0.80         | 0.01             |
| 7    | ACIDITY                                 | mgKOH/g            | IEC 62021-1        | MAX 0.01         | 0.00             |
| 8    | APPEARANCE                              | -                  | VISUAL             | BRIGHT AND CLEAR | BRIGHT AND CLEAR |
| 9    | BREAKDOWN VOLTAGE                       | kV                 | ASTM D877          | 30 - 999         | 48               |
| 10   | CORROSIVE SULFUR                        | -                  | DIN 51353          | NEGATIVE         | NEGATIVE         |
| 11   | QUANTITATIVE DIBENZYLDISULFIDE (DBDS)   | mg/kg              | IS 16497-1         | MAX 5.000        | 0.000            |
| 12   | DIELECTRIC DISSIPATION FACTOR           | -                  | IS 16086           | MAX 0.005        | 0.002            |
| 13   | DENSITY AT 20°C                         | g/mL               | ASTM D7042         | 0.8000 - 0.8950  | 0.8353           |
| 14   | INHIBITOR                               | %Wt                | IS 13631           | MAX 0.010        | 0.010            |
| 15   | INTERFACIAL TENSION AT 27°C             | mN/m               | ASTM D971          | TO REPORT        | 42.0             |
| 16   | KINEMATIC VISCOSITY AT 0°C              | mm <sup>2</sup> /s | ASTM D445          | MAX 1800.00      | 56.94            |
| 17   | KINEMATIC VISCOSITY AT 40°C             | mm <sup>2</sup> /s | ASTM D7042         | MAX 15.00        | 10.15            |
| 18   | METAL PASSIVATOR ADDITIVES              | mg/kg              | IS 13631           | 1.00 - 5.00      | 1.00             |
| 19   | PARTICLE COUNT                          | -                  | IS 13236           | TO REPORT        | 5                |
| 20   | POLYCYCLIC AROMATICS                    | %Wt                | IP 346             | MAX 3.00         | 1.00             |
| 21   | POLYCHLORINATED BIPHENYLS               | mg/kg              | IS 16082           | MAX 2            | 0                |
| 22   | CORROSIVE SULFUR                        | -                  | DIN 51353          | NEGATIVE         | NEGATIVE         |
| 23   | POUR POINT                              | °C                 | ASTM D97           | MAX -10          | -15              |
| 24   | SULPHUR CONTENT                         | %Wt                | ASTM D6443         | MIN 0.000        | 0.002            |
| 25   | WATER CONTENT                           | mg/kg              | ASTM D95           | MAX 30           | 28               |

Remarks : The Sample meets the Specification in the above tested parameters only as per IS 335:2018.

**Tested By:**  
PARVEEN RUBANA (31989940)

**Reviewed and Authorised By:**  
PARVEEN RUBANA (31989940)

Note:  
1) The sample is drawn by client and results apply to the sample as received.  
2) The test report shall not be reproduced except in full without prior written approval of the Lab In-charge.  
3) This test report shall not be used in any advertising media or as evidence in the Court of Law without prior written consent of Laboratory.  
4) Test results reported are valid for the sample tested, which was provided at the time of testing.  
5) The details of sample submitted is provided by sender / customer. The authenticity / correctness of the submitted data is the responsibility of the sender/customer.  
6) This is a system generated test report. Hence, it does not require signature.

